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Nature and Origin of Hematite in the Moenkopi Formation (Triassic), Colorado Plateau: A Contribution to the Origin of Magnetism in Red Beds

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Petrographic studies show that hematite is present in the Moenkopi Formation in at least five and possibly six forms: (1) microcrystalline hematite, (2) crystals of specular hematite, (3) polycrystalline and monocrystalline grains, (4) grains of partly hematitized ilmenite, (5) grains composed of primary ilmenite-hematite intergrowths, and (6) ultrafine pigment. The microcrystalline hematite and crystals of specular hematite are unequivocally authigenic. They form cement in interstitial and secondary voids, and they have replaced detrital iron-bearing silicate minerals. Furthermore, microcrystalline hematite is superimposed on other authigenic cementing minerals such as potassium feldspar, calcite, dolomite, and quartz, and in some cases it has replaced authigenic pyrite. In addition, both microcrystalline and specularite crystals are common daughter products of intrastratally altered biotite grains. Thermodynamic considerations coupled with studies of hematite-magnetite relationships in modern sediments indicate that most of the hematite in the polycrystalline grains, and probably the monocrystalline grains as well, was formed authigenically by post-depositional replacement of detrital grains of magnetite. The ilmenite probably has similarly altered in situ to hematite. The only hematite of unquestionable detrital origin in the red beds is the hematite in the ilmenite-hematite intergrowths ('tiger striped' grains) and that in monocrystalline detrital grains containing rutile exsolution platelets, both of which are products of high-temperature processes. With the exception of the ultrafine pigment, each of the above forms is coarser grained than the superparamagnetic threshold for hematite, and therefore each contributes components of remanent magnetism to the rocks. Inasmuch as most of the hematite varieties represent authigenic products of intrastratal alterations that require considerable geologic time, we conclude that the red bed remanence is largely chemical remanent magnetization (CRM) acquired over long time intervals. The pigment in the Moenkopi red beds consists partly of authigenic ultrafine red iron oxide and partly of translucent microcrystalline hematite. The ultrafine red iron oxide may or may not be hematite, but even if it is, the grain size probably lies below the paramagnetic threshold for hematite. Much of the pigment, therefore, may not contribute greatly to the remanent magnetism in the rocks.

INTRODUCTION

Red beds generally are more strongly magnetized than other types of sedimentary rocks, and consequently, they have been used extensively in paleomagnetic studies. Many investigators are convinced that the remanence in red beds closely reflects the geomagnetic field at the time of deposition and that the remanence can provide, in addition to average field directions for use in paleopole calculations, a detailed record of such events as secular variation and polarity reversals [*Helsley,* 1969; *Helsley and Steiner,* 1974; *Khramov,* 1967; *Van Den Ende,* 1970]. Many authors assume that remanence acquisition in red beds is contemporaneous with the time of deposition, and they conclude that the sequence of polarity reversals can be used to establish time lines for use in local and long-range correlations and for determining chronological relationships between sedimentary facies [*Helsley,* 1969; *Helsley and Steiner,* 1974; *Shoemaker et al.,* 1973; *Shoemaker and Purucker,* 1974; *Steiner and Helsley,* 1974*a*, *b*; *Turner and Ixer,* 1977; *Elston and Purucker,* 1979]. The use of remanence data in this manner is termed red bed magnetostratigraphy.

On the other hand, abundant data available in the literature indicate that red beds tend to be variably affected by magnetization acquired subsequent to deposition, i.e., 'secondary' magnetization [*Chamalaun and Creer,* 1964; *Al-Khafaji and Vincenz,* 1971; *Roy and Park,* 1972; *Steiner and Helsley,* 1972; *McGlynn et al.,* 1974; *Hoblitt et al.,* 1974; *Larson and Walker,* 1975; *Turner and Archer,* 1975; *Johnson,* 1976; *Turner,* 1979]. The results of these studies fuel a healthy skepticism of red bed magnetostratigraphy. The skeptics consider it likely that the polarity zonation in any sequence of red beds is merely a fortuitous consequence of remagnetization and carries little data useful for stratigraphic purposes.

The time of emplacement of the remanence is the crucial issue, and therefore the heart of the controversy centers on the origin of the hematite (the principal magnetic mineral in red beds), which carries the remanance. Is the hematite detrital or authigenic? If detrital, were the grains deposited as hematite or as magnetite that later altered (martitization) to hematite?

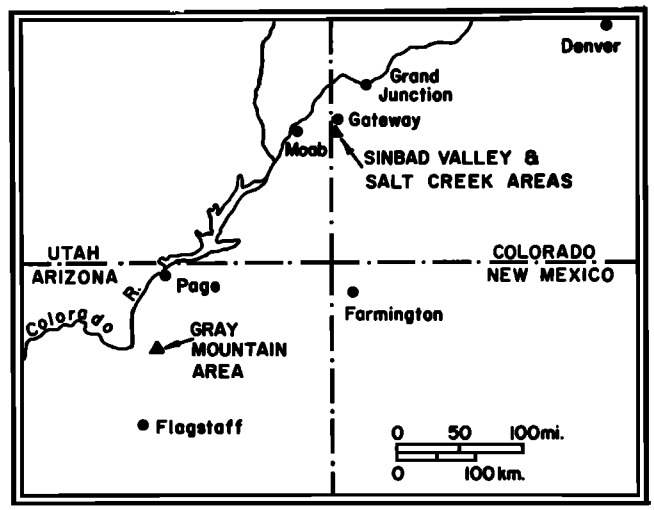


Fig. 1. Sampling localities.

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If authigenic, did the hematite form soon after or long after deposition?

These questions are not easily answered—which is why the controversy persists. The pros and cons of red bed magnetostratigraphy were the subject of a Penrose Conference in January 1978, but no consensus was achieved regarding the time of remanence emplacement. Neither side could demonstrate compelling evidence in its favor. To date, however, most magnetostratigraphic and paleomagnetic studies involving red beds have relied primarily on remanence measurements and rock magnetic studies. Very few studies have involved detailed petrographic observations. Petrographic studies, however, are useful in identifying the various mineral species present in the rocks and are essential for determining sequences of diagenetic changes, which may have affected the magnetic properties of the rocks.

During the past 4 years, we have combined magnetic measurements and petrographic observations in studies of samples of the Moenkopi Formation, a red bed unit that is widespread on the Colorado Plateau in western United States and one that has been considered by many to be the prime example of a well-behaved red bed magnetostratigraphic unit that acquired its remanence during or soon after deposition [*Helsley*, 1969; *Helsley and Shoemaker*, 1973; *Helsley and Steiner*, 1974; *Shoemaker et al.*, 1973; *Shoemaker and Purucker*, 1974; *Elston and Purucker*, 1979; *Shive et al.*, 1979]. The principal aim of the present study was to increase our understanding of how and when this rock unit acquired its remanence. This paper is a description of the results and a discussion of the implications of the part of the study involving the petrographic analyses, while referring only peripherally to the results of the paleomagnetic and rock magnetic studies, which are still in progress.

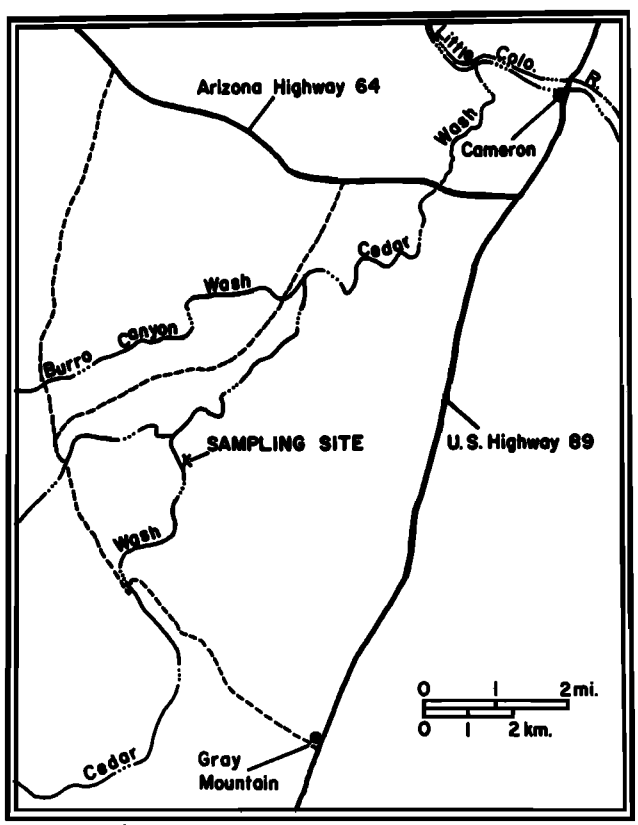


Fig. 2. Location of Gray Mountain, Arizona, sampling site.

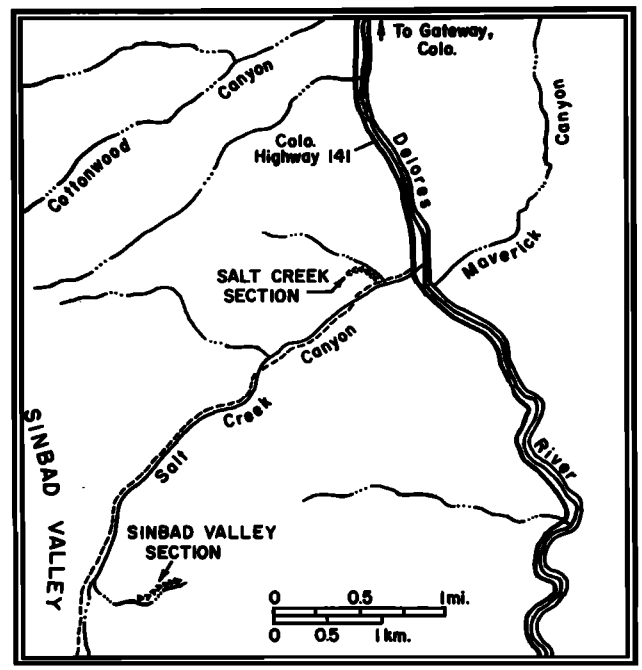


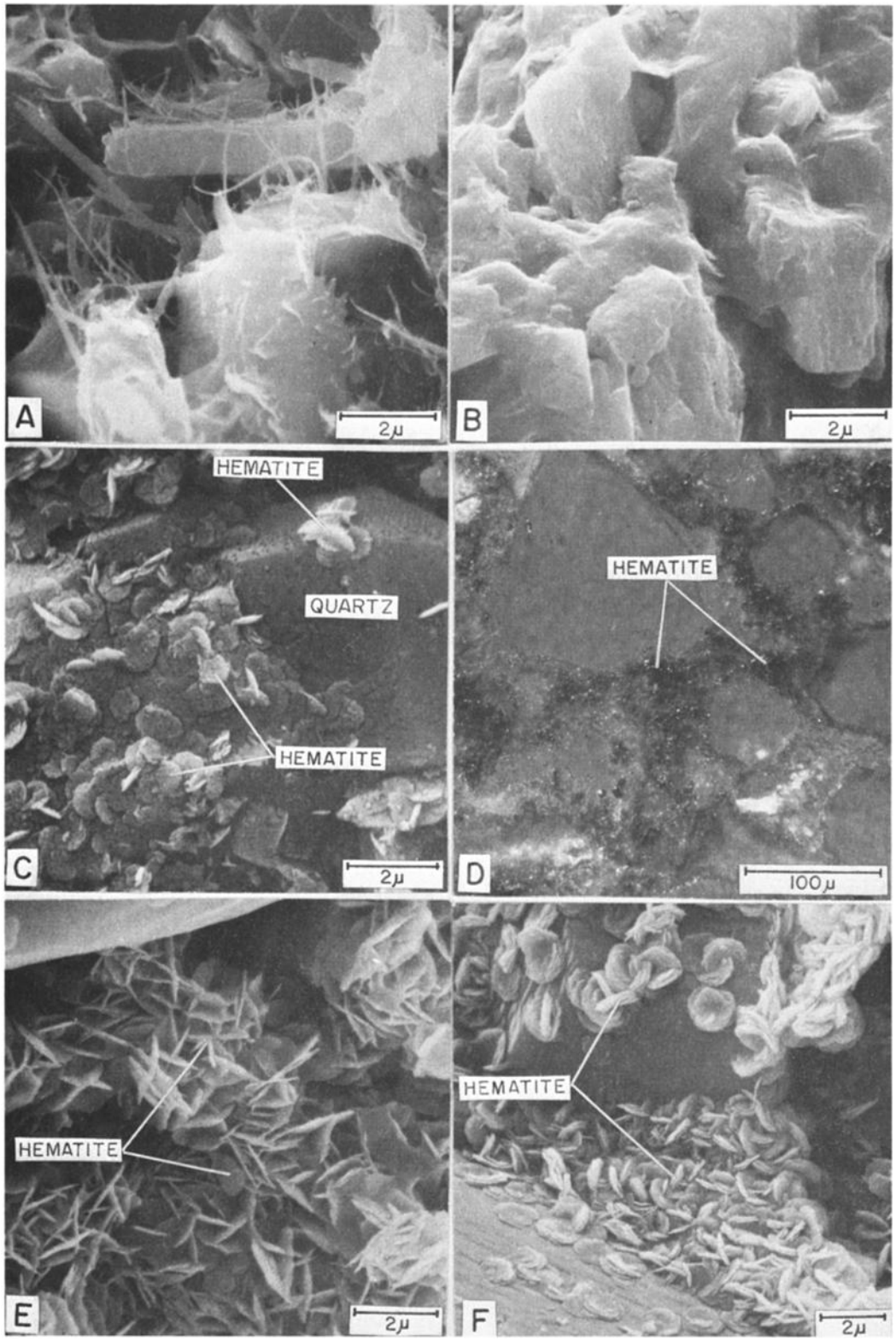
Fig. 3. Locations of Salt Creek and Sinbad Valley, Colorado, sampling sites.

PROCEDURE

Most of the samples used in this study are shale, siltstone, fine-to-coarse-grained sandstone, and pebbly sandstone, which vary in color from red, reddish brown, and pinkish through light gray. They were collected from outcrops in two widely separated areas in Arizona and Colorado (Figure 1). One area is along Cedar Wash, a tributary of the Little Colorado River, several kilometers northwest of Gray Mountain, Arizona (Figure 2); its exact location is SE quarter, section 14, T 28N, R 8E. The other area is located a few kilometers southwest of Gateway, Colorado (Figure 1), and includes out-

Fig. 4. (Opposite) (*a–b*) SEM photomicrographs of red matrix in a representative sample of very fine grained red sandstone. The particles in both photos are stained with red (10R $\frac{4}{6}$) iron oxide that shows no discernible crystal shapes when magnified 50,000×. The acicular to thread-like material in 4*a* is red-stained authigenic clay (probably illite). Figure 4*b* shows no evidence of similar clay authigenesis. Upper Ali Baba Member on east flank of Sinbad Valley, Colorado. (*c*) SEM photomicrograph showing microcrystalline hematite distributed more or less uniformly through interstitial matrix in a sample of bright red siltstone. A cluster of hematite crystals also encrusts the surface of a quartz overgrowth in the upper right-hand corner of the photo. Wupatki Member below the lower massive sandstone near Gray Mountain, Arizona. Sample has been etched in dilute hydrochloric acid to remove carbonate cement and to accentuate the hematite crystals. (*d*) Thin section photomicrograph showing blotches of authigenic hematite crystals that grade from bright red to black when viewed under inclined reflected light. An SEM photomicrograph of an analogous blotch from the same sample is shown in Figure 6*a*. Lower massive sandstone in Wupatki Member, near Gray Mountain, Arizona. (*e*) SEM photomicrograph showing a typical example of a meshwork of interlocking hematite crystals. The crystals line the walls of a sand-size void that probably represents the former location of an unstable iron-bearing mineral grain that has been intrastratally dissolved, leaving a residue of authigenic hematite. Lower massive sandstone in the Wupatki Member, near Gray Mountain, Arizona. (*f*) SEM photomicrograph showing typical microrosettes of authigenic hematite crystals concentrated along a heavy mineral lamina. Ali Baba Member on the east flank of Sinbad Valley, near Gateway, Colorado.

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crops near the mouth of Salt Creek (Figure 3) and on the east flank of Sinbad Valley (Figure 3). The magnetic properties of the red beds in the Gray Mountain area have been studied by *Shoemaker and Purucker* [1974] and *Elston and Purucker* [1979]; red beds in and near Salt Creek Canyon have been studied by *Helsley* [1969] and *Helsley and Steiner* [1974].

Twenty-eight samples from the Gray Mountain locality and 108 samples from the Salt Creek-Sinbad area were studied in thin and polished section at magnifications up to 1000×. In addition, selected samples that are representative of the common lithologies in each area were studied by means of a Cambridge Instruments scanning electron microscope (SEM) with attached Kevex X ray energy spectrometer (XES). The X ray spectrometer enabled us to obtain semiquantitative chemical

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analyses of material as small as one micron in diameter. In all cases, the chemical composition of minerals examined under the SEM were established by XES analysis. Mineralogy was determined petrographically and by X ray diffraction analysis (copper radiation). Identification of carbonate minerals was aided by staining the thin sections with alizerin red-S dye.

NATURE OF THE MAGNETIC IRON OXIDES

While there is much controversy concerning the timing of remanence acquisition in the Moenkopi Formation, all investigators agree that the stable remanence is carried primarily by hematite [*Helsley,* 1969; *Helsley and Steiner,* 1974; *Elston and Purucker,* 1979; this study] in grains coarser than 0.025 μm, the superparamagnetic threshold for hematite [*Banerjee,* 1971]. In addition, an indeterminant amount of remanence is carried by a tiny amount of detrital magnetite. Any effect due to magnetite, however, can be removed by heating the samples to temperatures above 580°C, the Curie temperature of pure magnetite.

Most of the samples are stained by a red iron oxide pigment, which appears to be either noncrystalline or crystalline on such an ultrafine scale that the crystals are not discernible by SEM analysis. Definitely identifiable hematite in these rocks consists of a microcrystalline variety and several varieties of specular hematite. The microcrystalline variety is bright red and translucent under transmitted light and contributes to the red color of the rocks. In contrast, the specular varieties are black in hand sample and opaque under transmitted light; they contribute little to the overall color.

Ultrafine Red Pigment

Most of the red pigment in the Moenkopi Formation is iron oxide that is either amorphous or too fine grained to show evidence of discernible crystals, even when magnified 50,000× (the highest magnification that can be obtained with reasonable resolution on the SEM at the University of Colorado). This type of iron oxide forms coatings on the framework and matrix grains and on cementing minerals. It appears to be the principal pigmenting material in all of the red samples examined. Examples of particles coated with the ultrafine pigment are shown in Figures 4*a* and 4*b*.

We do not know if this common pigment is hematite. Moreover, we do not know if, because of its fine grain size, it contributes to the magnetism of the rocks. X ray analyses of the <2-μm fraction from samples stained by typical ultrafine red pigment show the characteristic 33.3° 2θ peak of hematite. However, it may be that the hematite X ray peaks result from the presence in the samples of the microcrystalline variety of hematite described below. Even if the ultrafine pigment is hematite, its grain size appears to be below the superparamagnetic threshold [*Banerjee,* 1971], and it probably does not contribute greatly to the remanent magnetization of the red beds.

Microcrystalline Hematite

Microcrystalline hematite, as we define it, consists of subhedral to euhedral crystals that typically range from about ⅓ μm to about 2 μm in diameter (Figures 4*c*–4*f*, Figures 5 and 6). In some instances, individual crystals are scattered more or less uniformly through the matrix of the rock (Figure 4*c*). In most instances, however, microcrystalline hematite is locally concentrated in irregularly shaped blotches (Figure 4*d*). Such local concentrations are scattered throughout the rocks, but they are particularly abundant along heavy mineral laminae, and they commonly help to accentuate the laminae. The crystals are present in interlocking networks (Figure 4*e*), microrosettes (Figure 4*f*), and spherical to oblate clusters (Figures 5*a* and 5*b*). The crystals also form drusy coatings on the surfaces of framework grains (Figures 5*c* and 5*d*), and irregular aggregates that partly fill interstitial voids (Figures 5*c*, 5*e*, and 5*f*).

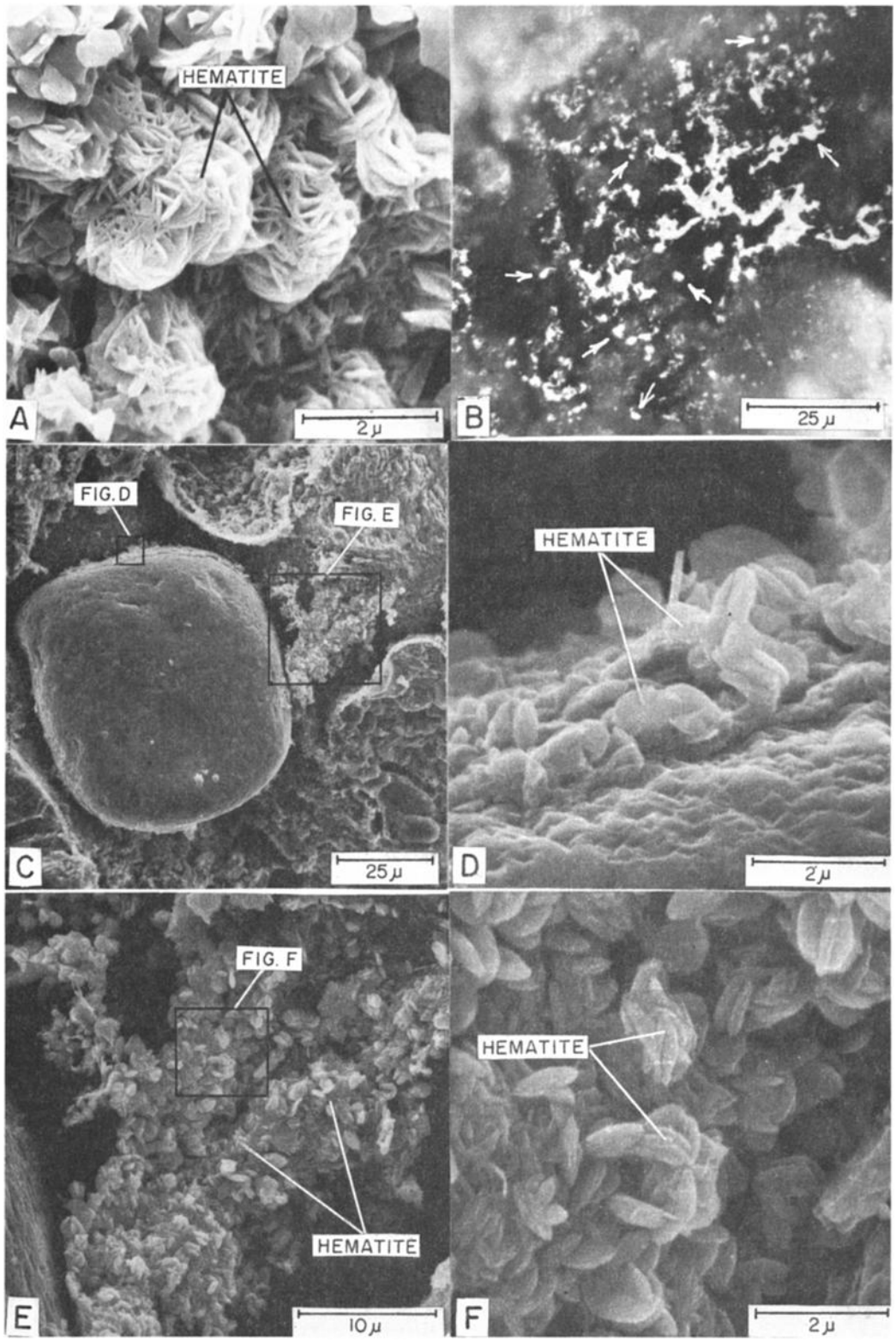
When examined in polished section under vertical illumination, with immersion oils and crossed nicols, the microcrystalline variety of hematite displays intense blood red internal reflections. In unpolarized light, large masses of microcrystalline hematite appear grayish-white where they intercept the polished surface and blood red around their margins below the intercepted surface. Single crystals and small aggregates (for example, rosettes and spherical and oblate clusters) are translucent and possess a vitreous luster that causes them to resemble blood red glass beads (Figure 5*b*).

When hand samples or thin sections that contain blotches of microcrystalline hematite are viewed under obliquely reflected light, the blothces commonly display gradations in color from bright red through darker reds to black. SEM studies show that the color gradations reflect either differences in grain size of individual hematite crystals or differences in the size and density of crystal aggregates (Figures 6*a*–6*d*). Small hematite crystals, such as those encrusting the large hematite crystal in Figure 6*a*, and those forming the interlocking meshwork in Figure 6*c*, invariably produce the brightest red colors. With increased crystal size (Figure 6*a*) and with increased size and density of the crystal clusters (Figure 6*d*), the colors in the blotches becomes darker red to black. The coarser-grained microcrystalline hematite (Figure 6*a*) is gradational into even larger-grained specular hematite (discussed below), which is opaque under transmitted light and moderately reflective (in hues of bluish white to grayish white) under vertical reflection.

Additionally, in most Moenkopi red beds, microcrystalline hematite is associated with altered biotite grains. In such occurrences, the hematite crystals characteristically are concentrated along cleavage lamellae, and where they are highly concentrated, they often cause the biotite flakes to be splayed apart, usually at the ends of grains. When such grains are split along cleavage lamellae, separated, and examined under the

Fig. 5. (Opposite) (*a*) SEM photomicrograph of spherical to oblate clusters of microcrystalline hematite. When viewed on a polished surface under vertically reflected illumination and crossed nicols (see 5*b*), these clusters resemble blood-red glass beads. The clusters are present separately and in aggregates, as shown in 5*b*. Lower massive sandstone, Wupatki Member, near Gray Mountain, Arizona. (*b*) Thin section photomicrograph of a polished surface under vertically reflected illumination, showing globular blebs (a few indicated by arrows) composed of spherical to oblate clusters of microcrystalline hematite similar to those shown in 5*a*. All of the white area represents globules, clusters, and branching networks of authigenic hematite. Same hand sample as in 5*a*. (*c*) SEM photomicrograph showing distribution of authigenic microcrystalline hematite in interstitial voids. Outlined areas are locations of 5*d* and 5*e*. Lower massive sandstone in Wupatki Member near Gray Mountain, Arizona. (*d*) SEM photomicrograph showing close-up view of microcrystalline hematite that forms the drusy coating on the framework quartz grain at the location shown in 5*c*. (*e*) Enlarged view of area outlined in 5*c*, showing an interstitial aggregate of microcrystalline hematite. (*f*) Enlarged view of area outlined in Figure *e*, showing subhedral nature of the hematite crystals.

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SEM, the cleavage surfaces are commonly found to be coated with microcrystalline hematite euhedra (Figures 6*e* and 6*f*).

All of the microcrystalline hematite is authigenic, and although the crystals typically are extremely fine grained, they are much larger than the superparamagnetic threshold size for hematite [*Banerjee,* 1971]. Therefore they should carry components of secondarily generated remanent magnetization. Moreover, all of these varieties of microcrystalline hematite contribute to the red pigment in the Moenkopi red beds.

Specular Hematite

Specular hematite is a relatively coarse-grained variety. Under transmitted illumination it is opaque, and under vertically reflected light (with immersion oils) it is moderately reflective

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in shades of bluish to grayish white. It is present in the Moenkopi red beds in four common forms: (1) individual crystals and crystal aggregates (including pods formed along cleavage planes in biotite), (2) detrital grains composed of polycrystalline and monocrystalline hematite, most of which has formed from partial to complete replacement (martite pseudomorphs) of magnetite grains, (3) detrital grains consisting of oxidized ilmenite, and (4) detrital grains composed of primary intergrowths of hematite and ilmenite.

Individual crystals and crystal aggregates. Individual euhedral specularite crystals typically range from a few microns to several tens of microns in diameter. Thus they are many times larger than the microcrystalline variety just described. They tend to be concentrated locally in clusters, most commonly along heavy mineral laminae, but they are also distributed more or less randomly through the rock material. Locally, the crystals form interstitial cement (Figure 7*a*). They are also concentrated in areas of about the same size as associated framework grains (Figure 7*b*). In the latter occurrence it appears that the specular hematite is the product of precipitation of iron released by intrastratal alteration of pre-existing unstable iron-bearing detrital materials. In some instances, clearly discernible outlines of what must have been former framework grains that have been dissolved in situ are marked by rims of authigenic specular hematite and (or) microcrystalline hematite (Figures 7*c*–7*e*). Moreover, clusters composed of mixtures of (1) specular hematite crystals, (2) microcrystalline hematite, and (3) associated iron oxide-stained clay minerals, all of which are likely products of intrastratally altered iron-bearing detrital silicate minerals, are frequent constitutents of the heavy mineral fractions of the red beds (Figure 8*a*). In rare instances, relics of what seem to be the parent silicate mineral still remain in the clusters (Figures 8*a* and 8*b*). In most instances, however, the parent silicate minerals have been removed completely, and the mixtures consist only of authigenic hematite and clay minerals. When viewed under vertically reflected light on polished surfaces, the locations of grains that have been altered in this manner are revealed by local concentrations of crystals of microcrystalline and specular hematite associated with nonreflective clay (Figure 8*c*).

Crystals of specular hematite are also commonly associated with altered biotite grains. The crystals, generally in pod-like bodies, are typically developed between cleavage lamellae, and under vertically reflected light (with immersion oils) they appear white to light gray (Figure 8*d*). In some cases they are associated with the finer-grained microcrystalline hematite (Figure 6*e*); in others, crystals of authigenic specular hematite, in addition to occurring as intercleavage pods, encrust the surfaces of biotite grains (Figure 8*e*).

Still another common occurrence of authigenic crystals of specular hematite is as overgrowths on detrital iron oxide grains (formerly magnetite) that are now composed of hematite (see below). The crystals are sharply euhedral, and they are in optical continuity with the hematite on which they grow (Figures 9*a* and 9*b*). Inasmuch as transportation would round off such protuberances, their presence confirms that the overgrowths grew in situ. Analogous occurrences of authigenic hematite crystals developed as overgrowths on detrital iron oxide grains have been reported in other red beds by *Turner and Ixer* [1977] and *Ixer et al.* [1979].

Detrital grains. Detrital magnetic oxide grains are present in every Moenkopi sample. The amount varies from about 0.1 to 0.3 volume percent in most samples to as high as 3 volume percent in samples containing several heavy mineral laminae. Four principal forms are present: (1) polycrystalline hematite (martite), (2) monocrystalline hematite, (3) hematite altered from ilmenite, and (4) primary ilmenite-hematite intergrowths.

Polycrystalline hematite (martite): About two-thirds of the detrital opaque grains are now composed of polycrystalline specular hematite. The evidence given below indicates unequivocally that this form has pseudomorphically replaced preexisting magnetite. It is properly called martite [*Deer et al.*, 1966].

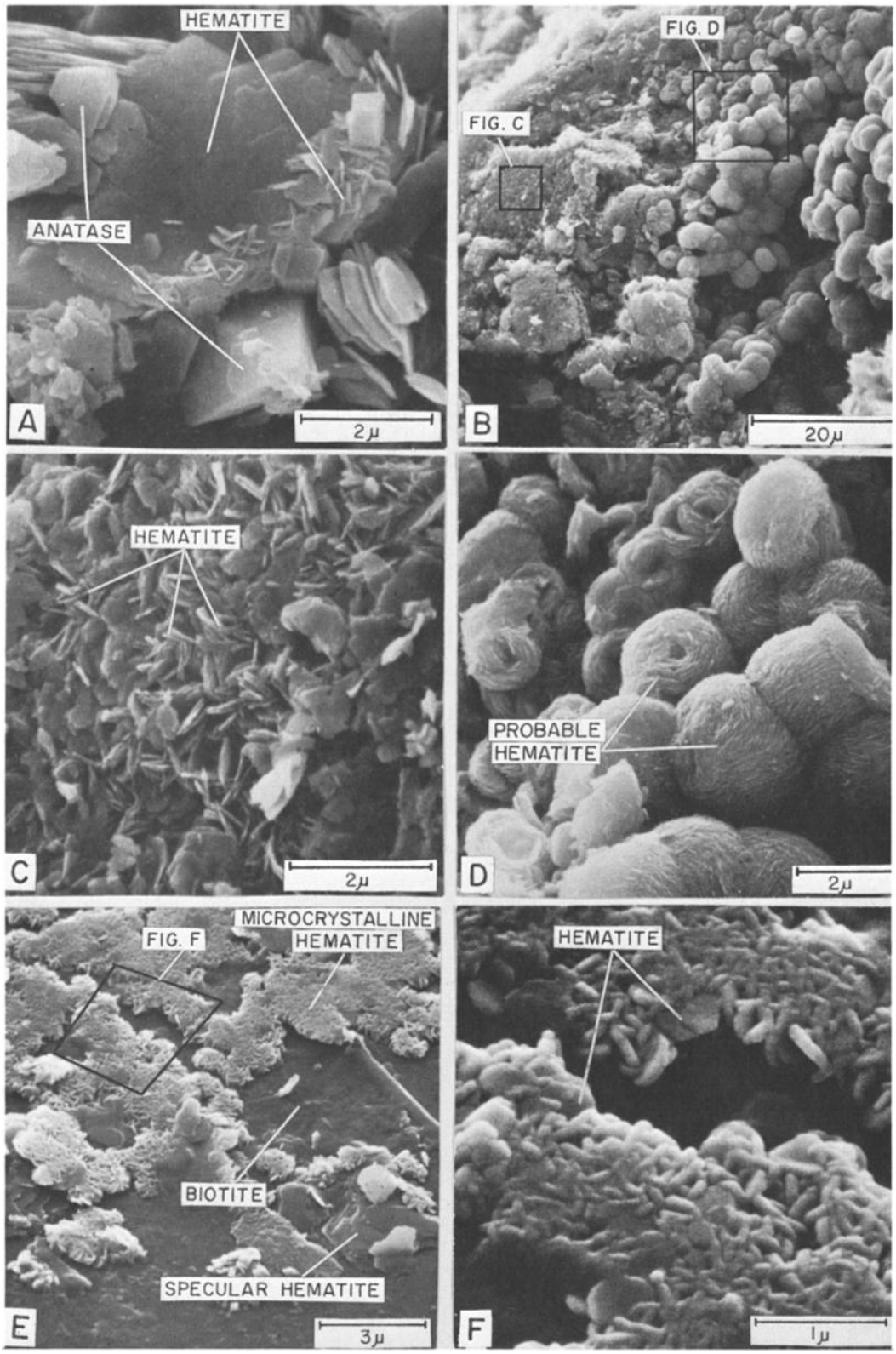
In most grains, martitization has been so extensive that none of the original magnetite remains. Indeed, relics of the parent magnetite persist in only about 2–3% of the martitized grains, and within these grains the relict magnetite never occupies more than about 2–3 volume percent. The martite grains, when viewed under crossed nicols using vertical illumination and immersion oils, are seen to consist of complex intergrowths of hematite. Some of the intergrowths have clearly formed by replacement along the octohedral (111) planes of the preexisting magnetite (Figure 9*c*). Other grains display mosaics of interlocking, irregularly shaped hematite crystals that presumably reflect simultaneous growth from multiple centers of replacement (Figure 9*d*).

It is noteworthy that none of the martite grains have been observed to contain ilmenite lamellae disposed along the (111) planes of the preexisting magnetite. The lack of such lamellae is an indication that the magnetite initially contained only small amounts of titanium and was likely derived from felsic plutonic and metamorphic source rocks [*Buddington and Lindsley*, 1964].

Martite grains that are coarse enough to be seen with a petrographic microscope are much coarser grained than the superparamagnetic threshold size for hematite, and therefore the grains should carry a stable remanence. However, whether that remanence is a true detrital remanent magnetism (DRM) or a CRM is an open question. The crucial point concerns the time of alteration of magnetite to hematite. If it took place in the source area, then the grains would already be hematitized when deposited and would probably carry a DRM component that closely reflects the geomagnetic field at the time of deposition. If, however, martitization was primar-

Fig. 6. (Opposite) (*a*) SEM photomicrograph of authigenic hematite and associated authigenic anatase localized in concentrations similar to those shown in Figure 4*d*. The tiny crystals are bright red; the coarse crystals are dark red. Lower massive sandstone in Wupatki Member near Gray Mountain, Arizona. (*b*) SEM photomicrograph of a local concentrate of microcrystalline hematite that grades from bright red to nearly black. The bright red colors are produced by the interlocking meshwork of crystals shown in the left-hand side of the photo and enlarged in 6*c*. The nearly black colors are produced by the dense, ball-like aggregates of crystals (probably hematite) shown in the right-hand side of the photo and enlarged in 6*d*. X ray energy spectrometer analysis show that both types of crystals are similarly composed of pure iron oxides. Upper Ali Baba Member on east flank of Sinbad Valley. (*c*) Enlargement of area outlined in 6*b*, showing details of the interlocking meshwork of hematite crystals. The individual crystals are subhedral to euhedral hexagonal platelets. (*d*) Enlargement of area outlined in 6*b*, showing details of the ball-like aggregates of crystals. (*e*) SEM photomicrograph of the internal surface of a biotite grain that has been split apart along cleavage lamellae, exposing the intercleavage concentrates of microcrystalline hematite and occasional crystals of specular hematite. The outlined area shows the location of Figure 6*f*. Lower Moenkopi at the mouth of Salt Creek near Gateway, Colorado. (*f*) Enlarged view of area outlined in 6*e*, showing euhedral nature of the microcrystalline hematite.

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ily a product of diagenesis after deposition, then the component of remanent magnetism they carry may have been acquired long after deposition and may have no relationship to the earth's field at the time of deposition. We believe the following evidence indicates that most of the hematitization took place after deposition.

It is known that magnetite tends to be replaced by hematite whenever the Eh-pH of interstitial waters lies in the stability field of hematite [*Garrels and Christ*, 1965, see Figure 7.6]. Moreover, since red beds contain abundant hematite of all types, including martite, we have clear evidence that the Eh-pH of the interstitial waters was in the appropriate field. It fol-

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lows, therefore, that interstitial environments in red beds are particularly favorable for in situ martitization of magnetite. Indeed, martitization after deposition should be expected in all red beds in which magnetite was an original mineral.

Soils also commonly have Eh-pH conditions favorable for martitization, and therefore they can be important sources of detrital martite in sediments. To evaluate this possibility, *Van Houten* [1968] investigated the nature and origin of detrital grains of iron oxide minerals in a variety of igneous and metamorphic source rocks, upland soils, alluvium in transit, and Recent sediments from many places throughout the world. Van Houten found that regardless of the intensity of weathering in the source areas, most magnetite is delivered to depositional basins in a relatively unaltered condition. Consequently, Van Houten argued that grains of martitized magnetite in red beds are principally products of post-depositional oxidation of detrital magnetite. Moreover, Van Houten contended that the oxidation continues long after burial, since successively older red beds contain a proportionally larger percentage of martitized grains.

We have obtained additional evidence from Recent alluvium in both humid and arid regions that supports Van Houten's conclusions. Our evidence is based on petrographic examination of magnetite grains in Recent alluvium from (1) four rivers on the island of Puerto Rico (four samples), where the average annual rainfall in the source area exceeds 250 cm and the watersheds are characterized by deeply weathered lateritic soils, (2) arroyos and alluvial fans in northeastern Baja California (four samples), where the average annual rainfall is less than 10 cm, and (3) arroyos of the Sinai Peninsula (ten samples), where the average rainfall is about 5 cm/yr.

Magnetite grains in the alluvium from Puerto Rico display an average alteration to hematite of only 34 volume percent, and many grains are only slightly (less than 10%) altered. Alteration of magnetite to hematite in the alluvium in the two desert regions averages about 40%, and again, many grains are essentially unaltered.

Therefore on the basis of Van Houten's data, coupled with our own, we conclude that complete martitization of such a high percentage of the magnetite as occurs in the Moenkopi Formation (more than 98%) cannot be accounted for by source area weathering alone, under any type of cilmate. It is much more likely that most of the replacement took place after deposition, as the grains were bathed by interstitial waters in which the Eh-pH conditions were in the stability field for hematite. It also would seem reasonable that under these same conditions, intrastratal breakdown of unstable iron-bearing silicates would occur. One would also expect that penecontemporaneous with this breakdown, the ultrafine pigment, microcrystalline hematite, and crystals of specular hematite would also be precipitated.

The origin and character of the remanence carried by the martite grains is a particularly difficult problem to resolve. Those grains that had undergone complete alteration to hematite prior to deposition, albeit a small percentage of the total, would be capable of acquiring a true DRM at or close to the time of deposition—provided that the force couple on the grains was adequate to overcome inertial forces and bring about physical alignment. For hematite, which is about 100 times less magnetic than magnetite, the critical grain size below which alignment would occur would be much smaller than it is for magnetite. This would mean that those grains of hematite greater than about silt size when deposited would be largely unaffected by field alignment forces, particularly when deposited under conditions of turbulent flow.

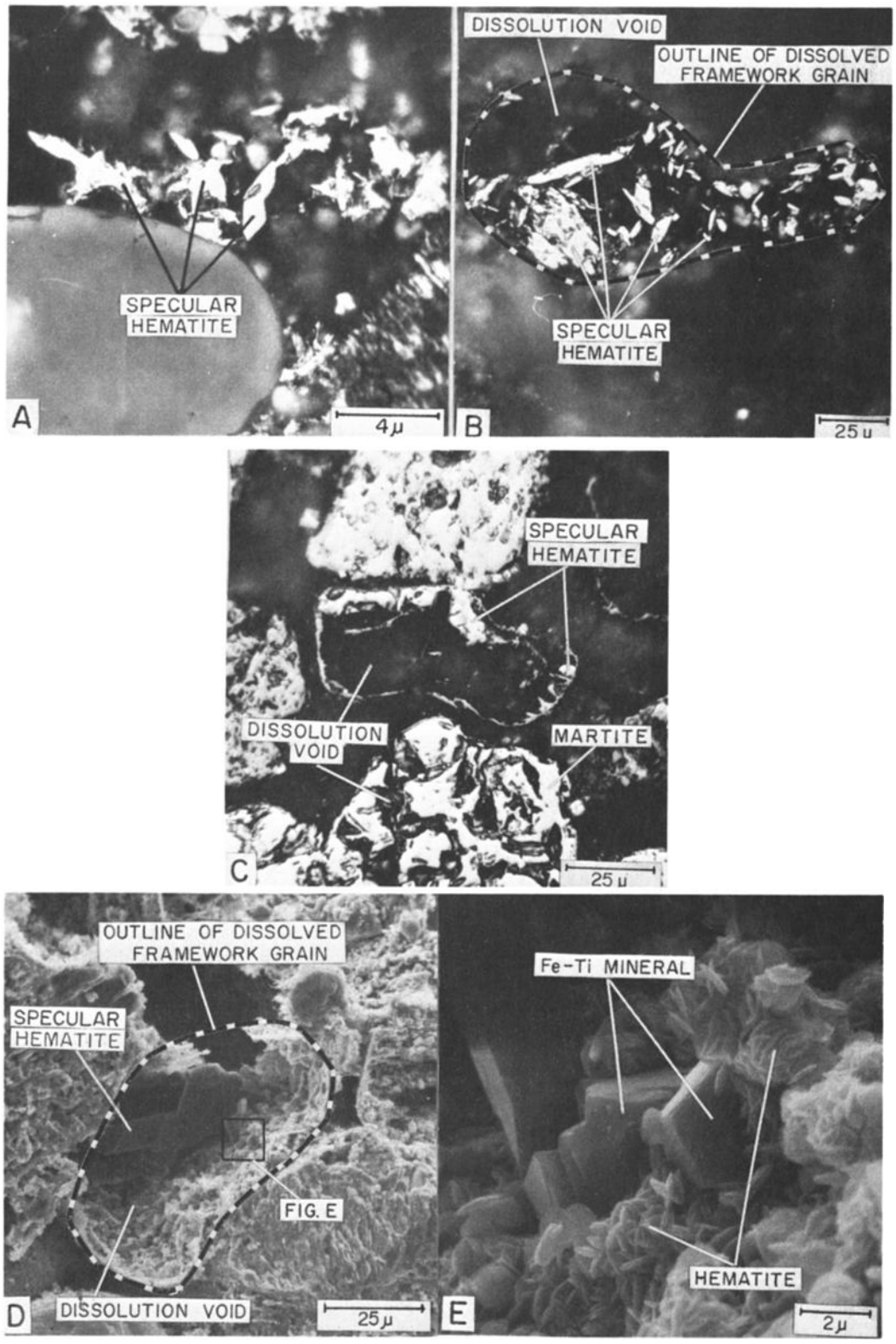
On the other hand, if, as we believe, martitization occurred largely subsequent to burial, then the remanence would principally be a CRM. In this case, however, the following question arises: will the remanence direction in the hematite mirror that of the magnetite or take on the direction of the local ambient geomagnetic field at the time of transformation from a cubic phase (magnetite) to a rhombohedral phase (hematite). D. J. Dunlop (written personal communication, 1979) is of the opinion that the crystallographic change is so great that spin coupling will be lost across the developing boundaries, and the newly formed hematite will acquire the direction of the ambient field. Recent laboratory experiments involving the alteration of magnetite to hematite in highly oxidizing solutions supports Dunlop's conclusion [*Amerigian*, 1978]. This is an important result, and this line of investigation merits further effort to settle the question conclusively.

Another interesting facet of the potential changes in remanence brought about by post-depositional alteration of magnetite is a consequence of their source. As previously noted, essentially all martite grains in the Moenkopi red beds lack ilmenite lamellae. The magnetite grains therefore would have been monocrystalline, and accordingly, they would have been magnetically soft in grains larger than about 30–40 μm. Following deposition, such grains would tend to acquire large components of VRM, regardless of the initial DRM direction. Consequently, in sediments containing a preponderance of detrital opaque grains larger than 30–40 μm, even if the remanence directions in the martite do mirror those in the magnetite, the final remanence direction appears likely to depend on the ambient field direction at the time the martitization occurred, rather than the initial DRM direction. This possibility also needs further study.

Monocrystalline hematite: Less than 10% of the detrital opaque grains are monocrystalline hematite; that is, they lack the characteristic interpenetrating polycrystalline pattern of most martite grains (such as those shown in Figures 9*c*–9*d*). Monocrystalline hematite grains can form as fragments derived from preexisting larger monocrystalline and coarse-grained polycrystalline masses of hematite. About one-third

Fig. 7. (Opposite) (*a*) Polished surface photomicrograph under vertically reflected illumination. White areas are the moderately reflective polished surfaces of euhedral crystals of authigenic specular hematite. The crystals have partly filled an interstitial void space along a heavy mineral lamina. Ali Baba Member on the east flank of Sinbad Valley. (*b*) Polished surface photomicrograph under vertically reflected illumination. White areas are crystals of specular hematite. The crystals are concentrated in a void created by dissolution of an unstable framework grain, probably a ferromagnesian silicate such as hornblende. Wupatki Member, below the lower massive sandstone, near Gray Mountain, Arizona. (*c*) Polished surface photomicrograph under vertically reflected illumination. All white areas are specular hematite. The photo shows a typical dissolved framework grain within a heavy mineral lamina. The original outline of the dissolved grain is marked by a rim of specular and microcrystalline hematite analogous to the rim of hematite surrounding the dissolution void shown in 7*d* and 7*e*. The other white grains are martite, one of which contains prominent internal dissolution voids. Lower massive sandstone in Wupatki Member near Gray Mountain, Arizona. (*d*) SEM photomicrograph of a dissolution void analogous to that shown in 7*c*. The outlined area within the void shows the area of 7*e*. Same hand sample as in 7*c*. (*e*) SEM photomicrograph of area outlined in 7*d*, showing drusy lining of microcrystalline hematite and an unidentified iron-titanium mineral.

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of the monocrystalline grains display exsolution plates of rutile ('blitz' texture of *Ramdohr* [1969]), which result from high-temperature magmatic (possibly metamorphic) processes [*Ramdohr*, 1969]. These grains clearly seem to be composed of primary hematite derived from crystalline source rocks. Monocrystalline grains can also form through pseudomorphic replacement of magnetite, just as in the case of the previously described polycrystalline hematite grains. Although hematite replacement of magnetite in most cases proceeds along the (111) planes, it can also sweep across the grains along a single front, producing single-crystal replacement (*Haggerty* [1976] and our own observations). Because of these differences in origin, some of the monocrystalline specularite grains may carry a true DRM (assuming they could be aligned by the magnetic

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field), whereas others, if altered post-depositionally, may carry a CRM. In any case, the proportion of monocrystalline specularite in the Moenkopi red beds is so small in comparison to that of polycrystalline hematite grains (less than 10% compared to 65–70%) that the remanence component contributed by it is minimal.

Hematite in oxidized ilmenite grains: Ten to twenty percent of all detrital opaque grains in the Moenkopi red beds consist of ilmenite and its alteration products. Altered grains outnumber unaltered grains by about 2:1. The alteration products are leucoxene/anatase and a blotchy two-phase intergrowth composed of hematite and a phase that most likely is iron-titanium oxide (Figure 9*e*). Under vertically reflected light the hematite appears white, whereas the iron-titanium oxide(?) phase is dark gray to purplish gray. In some grains the scattered irregularly shaped hematite patches are 1–2 μm across; in others they are as much as 20 μm across.

To our knowledge, intergrowths such as these have not been previously described. They are unlike intergrowths resulting from high-temperature processes [*Ramdohr,* 1969; *Haggerty,* 1976; *Lindsley,* 1976; our own observations]. Therefore we interpret them to be the result of low-temperature oxidation during diagenesis. We presume, but we cannot be certain, that the hematite phases of these intergrowths carry a CRM.

Ilmenite-hematite intergrowths: Grains consisting of primary intergrowths of ilmenite and hematite (which we term 'tiger striped' grains) make up between 5 % and 6 % of the detrital opaque grains in the Moenkopi red beds (Figure 9*f*). Such intergrowths are the result of subsolidus exsolution of a high-temperature solid solution of hematite and ilmenite into two separate phases primarily during slow cooling in grantitic plutons. The plane of the 'tiger striping' parallels the basal crystallographic plane, which is the easy direction of remanence [*Nagata,* 1961]. All tiger striped grains that are coarse enough to be identified under the microscope should be capably of carrying a remanence, and since the hematite in these grains is primary, they probably contribute to a true DRM (or PDRM). It is certain, however, that such grains, because they are composed of about 25–75% nonmagnetic ilmenite, will be even less susceptible to physical alignment in a magnetic field than are those of hematite. Physical alignment of weakly magnetic hematite grains in response to the earth's field is difficult at best; with the additional mass of the ilmenite, the efficiency of the alignment is further reduced. Moreover, in the case of tiger striped grains, it is not known whether the remanence in the alternate hematite bands is in parallel or antiparallel alignment. If the latter, then the effective remanence will be reduced nearly to zero. As far as we know, however, no one has conducted any research concerning the magnetic properties of ilmenite-hematite intergrowths.

In view of the small percentage of these grains present, their relatively weak remanence, and their high content of nonmagnetic ilmenite, it is likely that tiger striped grains contribute only a minimal amount to the total remanence in red beds.

EVIDENCE CONCERNING THE TIME OF HEMATITE FORMATION

Our evidence indicates that most hematite in the Moenkopi Formation has formed authigenically, after deposition of the sediments. It seems likely, therefore, that most of the magnetic remanence in these rocks is carried by the various species of authigenic hematite described above. The question of the timing of the authigenesis still remains: Did it occur shortly after or long after deposition? This question, so crucial to the understanding of red bed magnetism, is not easily answered. On the basis of the remanence record, *Helsley* [1969], *Helsley and Steiner* [1974], and *Baag and Helsley* [1974] have concluded that remanence is acquired penecontemporaneously with deposition. Others [*Roy and Park,* 1972; *Larson and Walker,* 1975; *Turner and Archer,* 1975; *Turner,* 1979; *Ixer et al.,* 1979] have contended that the authigenesis takes place long after deposition and may occur repeatedly.

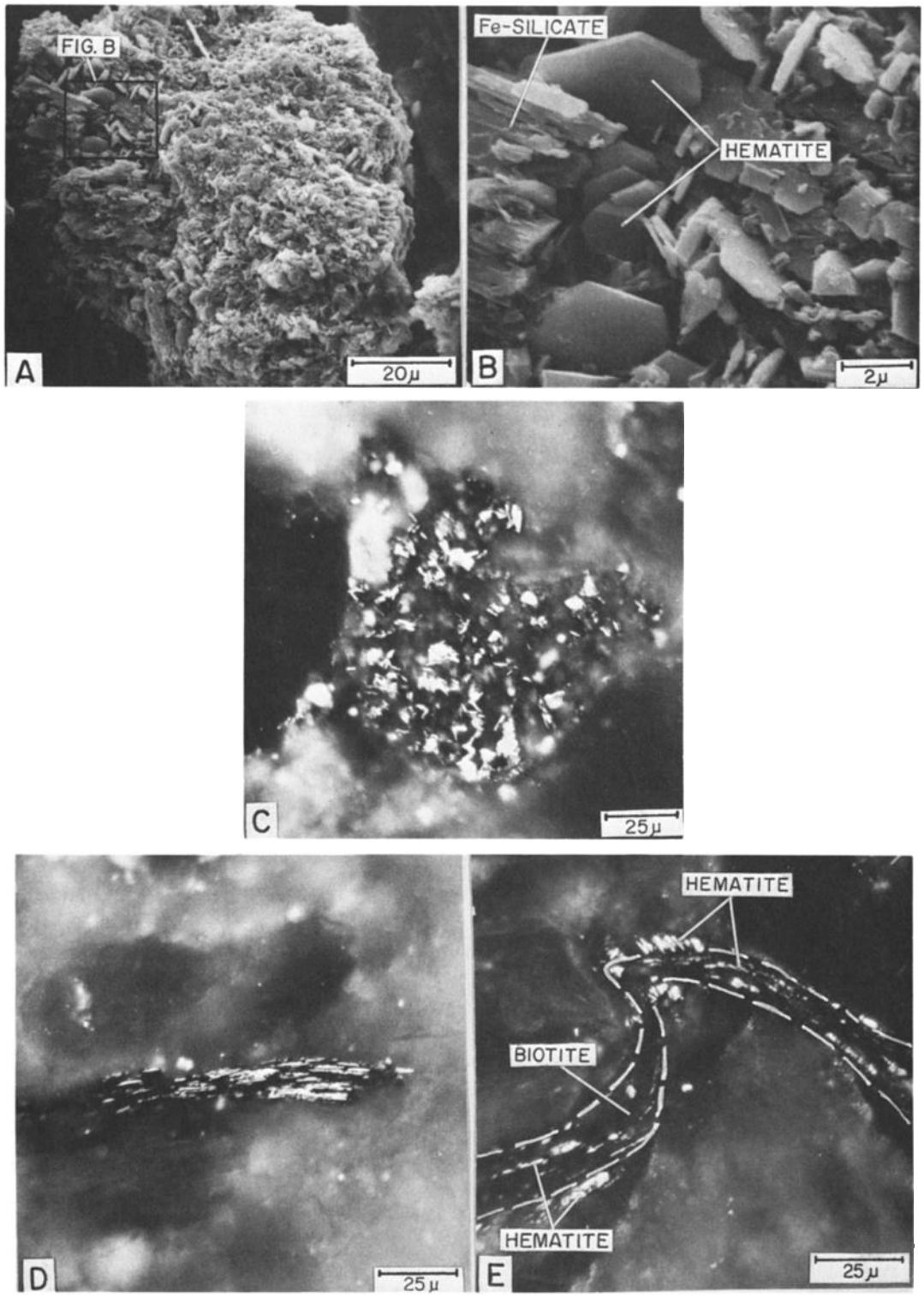
We believe the following lines of evidence indicate that the authigenesis of hematite (and remanence acquisition) took place long after deposition.

1. Lack of hematite in Holocene beds: If, indeed, hematite forms penecontemporaneously with deposition, one would expect to find examples of red Holocene deposits that show at least incipient stages of the formation of crystalline hematite. To our knowledge, no such deposits are known. We interpret the lack of Holocene examples, coupled with the commonplace occurrence of such hematite in older deposits, as a priori evidence that the formation of hematite is a geologically slow process. This reasoning is supported by ample petrographic evidence (discussed below) indicating that the formation of hematite has occurred subsequent to other types of diagenetic mineral alterations that also require the lapse of considerable geologic time.

2. Coatings on authigenic cementing minerals: Hematite and its precursor oxides are present in some samples of the Moenkopi as coatings on authigenic overgrowths on detrital framework grains such as potassium feldspar (Figures 10*a* and 10*c*) and quartz (Figure 4*c*). They are also present within crystals and on the crystal surfaces of interstitial cementing minerals such as dolomite and calcite (Figure 10*b* and 10*c*). In some instances, layering within overgrowths provides excellent evidence that iron oxides have been deposited repeatedly during the diagenetic history of the sediments. A remarkable example demonstrating episodic deposition is shown in Figure 10*c*. Apparently, the detrital feldspar grain (DF) shown in this illustration was either deposited with an iron oxide coating

Fig. 8. (Opposite) (*a*) SEM photomicrograph of a grain composed mainly of hematite crystals and clay. The grain is part of the heavy mineral fraction that was separated from a sample of the lower massive sandstone in the Wupatki Member near Gray Mountain, Arizona. We interpret this grain to be the product of in situ alteration of an unstable iron-bearing detrital silicate mineral such as hornblende. Outlined area shows location of 8*b*. See 8*c* for characteristics of an analogous grain viewed in polished thin section. (*b*) Enlarged view of the area outlined in 8*a*, showing crystals of hematite and a small relic of the highly altered parent iron-bearing silicate mineral (composition determined by XES analysis). (*c*) Polished surface photomicrograph under vertically reflected illumination, showing characteristics of a grain analogous to that illustrated in 8*a*, when viewed in situ in a polished thin section. All white areas are crystals and aggregates of crystals of specular hematite. Note numerous elongate cross sections of euhedral crystals. Lower massive sandstone in Wupatki Member near Gray Mountain, Arizona. (*d*) Polished surface photomicrograph under vertically reflected illumination, showing biotite grain that has been pervasively replaced by specular hematite. Lower massive sandstone in Wupatki Member near Gray Mountain, Arizona. (*e*) Polished surface photomicrograph under vertically reflected illumination, showing deformed and altered biotite grain displaying hematite between cleavage lamellae and in crystal aggregates encrusting the surface of the grain. Lower Moenkopi, near mouth of Salt Creek near Gateway, Colorado.

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(H-1) or, alternatively, it acquired an iron oxide coating during early diagenesis. Subsequently, the feldspar grain developed an authigenic overgrowth (FO), and relics of the preexisting iron oxide (H-1) were preserved at the grain-overgrowth interface. At some indeterminable later time, feldspar precipitation ceased, and iron oxide (H-2) was deposited on the surface of the overgrowth. Dolomite (D) was formed next, and it, in turn, acquired a later coating of iron oxide (H-3). The dolomite crystal developed two incomplete overgrowths, each of which was coated with still younger iron oxide (H-4 and H-5). Subsequently, dolomite precipitation ceased, and botryoidal calcite (C) was precipitated on the iron oxide-coated dolomite. Finally, the calcite was coated with a still younger stage of iron oxide (H-6).

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Examples such as these confirm that there have been multiple episodes of iron oxide deposition in the Moenkopi red beds. Moreover, the association of iron oxide with different types of cementing minerals, each of which reflects significantly different ground water chemistry and must therefore require long intervals of geologic time for formation, constitutes strong evidence that hematite has been sporadically deposited in these rocks long after deposition of the sediments.

3. Drusy linings of voids formed by dissolution of labile minerals: The fact that hematite crystals commonly form drusy linings in secondary voids (Figures 7*b–e*) is evidence that this hematite postdates the time of formation of the voids. The shapes and sizes of the voids are similar to the shapes and sizes of the surrounding framework grains. This fact constitutes strong evidence that the voids represent dissolved framework grains. For reasons that we will explain in the following section, we believe that many of the dissolved grains were ferromagnesian silicate minerals (such as hornblende) which, prior to their dissolution, were peripherally coated or (and) peripherally replaced by hematite. Presumably because of their chemical instability, the minerals subsequently were completely dissolved by groundwater, leaving the rims of hematite to delineate the original grain margins (Figure 7*c*). It is likely that the formation of hematite druses in such voids reflects considerable geologic time, inasmuch as the silicate mineral grains had to be dissolved before the drusy linings of hematite could be precipitated.

In our opinion, these voids are end products of processes of intrastratal dissolution similar to those that have partially dissolved hornblende and augite from some red beds of Cenozoic age in Baja California and southwestern United States [*Walker*, 1976; *Walker et al.*, 1978]. Inasmuch as the Cenozoic red beds display evidence of only partial removal of the hornblende and augite in rocks that are known to be of Miocene age, we conclude by analogy that complete dissolution of grains of these minerals must require millions of years, even under the most favorable conditions.

4. Hematitized pyrite: In some places on the east flank of Sinbad Valley (Figure 3), outcrops of Moenkopi red beds contain abundant hematite pseudomorphs of pyrite crystals. The pseudomorphs are concentrated along bedding planes on the lower surfaces of sandstone beds and range in size from less than 1 mm to about 4 mm. Samples containing the pseudomorphs have a reversed Triassic magnetic direction, which is the same as that in the red beds in which they occur. Samples containing the pseudomorphs, however, generally register about 10 times stronger intensity than samples lacking them, indicating that the hematite in the pseudomorphs is a prime carrier of remanence. We have mechanically demagnetized some of the samples by physically removing the pseudomorphs. The results definitely demonstrate that the pseudomorphs carry about 80–90% of the magnetization in the samples. Moreover, the Triassic direction provides direct evidence that the hematitization of the pyrite is a product of intrastratal alteration, not Holocene weathering.

It is not possible to determine exactly when the hematite pseudomorphs formed, but we can be certain that sufficient time lapsed following deposition to allow the formation of the large pyrite crystals, the change of the interstitial chemical environment from reducing to oxidizing, and the complete replacement of the pyrite by hematite.

5. Nearly complete hematitization of magnetite: The fact that grains containing relics of unaltered magnetite are common in Cenozoic red beds, less common in Mesozoic red beds, and rare in Paleozoic red beds [*Van Houten*, 1968; our own data] implies that nearly complete hematitization of magnetite, such as that seen in the Moenkopi red beds, is a product of intrastratal alteration that must have involved millions to even tens of millions of years. Inasmuch as thermodynamic data indicate that magnetite is unstable and alters to hematite in oxidizing environments that characterize red beds, it is likely that much of the replacement of the magnetite by hematite took place simultaneously with the in situ formation of the microcrystalline hematite and individual crystals and crystal aggregates of specular hematite.

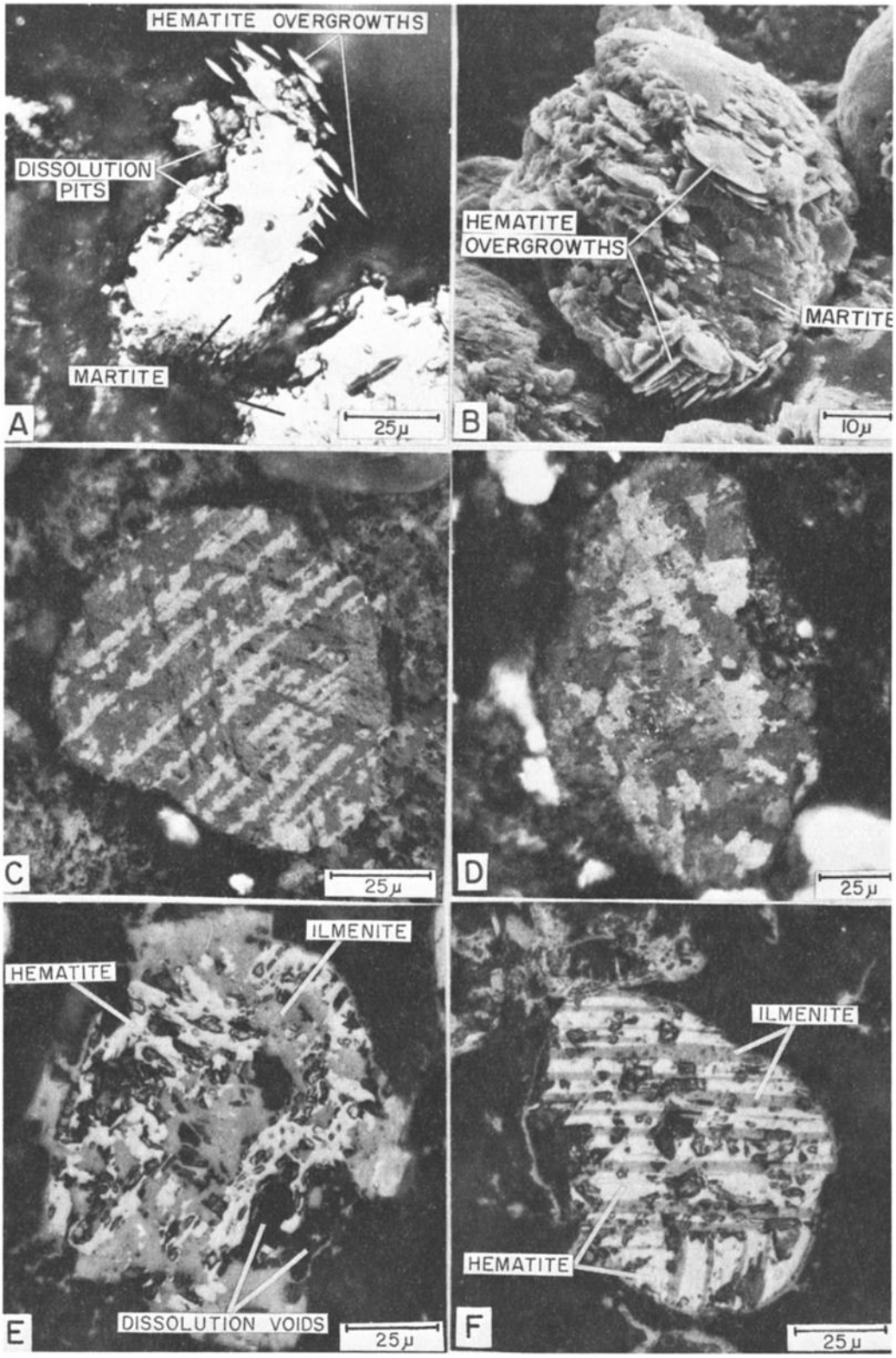
6. Two-phase alteration of ilmenite: As noted above, many of the detrital ilmenite grains have been altered to a blotchy two-phase association of hematite and an iron-titanium(?) mineral. Inasmuch as such intergrowths have not been reported in studies of opaque minerals associated with rocks formed by high-temperature processes, it is likely that they are the result of low-temperature (diagenetic) oxidation and diffusion, probably the same conditions that caused martitization of the detrital magnetite grains. Diffusion at low temperatures is a geologically slow process, and therefore if the intergrowths are indeed products of diagenetic processes, then a geologically long time interval is required for their formation.

SOURCES OF IRON CATIONS

There is a great deal of evidence indicating that the iron cations used in the development of ultrafine pigment, microcrystalline hematite crystals, and microcrystalline specular hematite, were principally derived from the breakdown, intrastratally, of iron-bearing detrital grains. The Moenkopi Formation at each of the two sampled localities contains abundant arkosic detritus derived from igneous and metamorphic

Fig. 9. (Opposite) (*a*) Martite grain displaying peripheral dissolution pitting and younger overgrowths of crystallographically continuous hematite crystals. Polished surface photomicrograph under vertically reflected illumination. Lower massive sandstone in Wupatki Member near Gray Mountain, Arizona. (*b*) SEM photomicrograph of a grain of martite with crystalline overgrowths of authigenic specular hematite. Lower massive sandstone in Wupatki Member near Gray Mountain, Arizona. (*c*) Polycrystalline hematite (martite) composed of a network of interpenetrating crystalline units that intersect along octohedral (111) planes inherited from the replaced parent magnetite grain. The entire grain is now hematite. The various shades of gray are caused by the effects of the different crystallographic orientations on the pleochroism. Polished surface photomicrograph under crossed nicols. Lower massive sandstone in Wupatki Member near Gray Mountain, Arizona. (*d*) Polycrystalline hematite (martite) composed of a mosaic of interlocking hematite crystals. The entire grain is hematite. The various shades of gray are caused by the effects of the different crystallographic orientations on the pleochroism. Polished surface photomicrograph under crossed nicols. Lower massive sandstone in Wupatki Member near Gray Mountain, Arizona. (*e*) Detrital grain of ilmenite (gray areas) that has been extensively altered to hematite (white areas). Black areas within the grain are dissolution voids. Polished surface photomicrograph under vertically reflected illumination. Lower massive sandstone in Wupatki Member near Gray Mountain, Arizona. (*f*) Detrital grain composed of 'tiger striped' ilmenite-hematite intergrowth. Gray layers (ilmenite) alternating with white layers (hematite) are products of true high-temperature exsolution in granitic plutons. The striping parallels the basal planes of hematite and ilmenite. Left side of grain has been pitted by dissolution. Polished surface photomicrograph under vertically reflected illumination. Lower massive sandstone in Wupatki Member near Gray Mountain, Arizona.

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source areas. Our petrographic studies, like earlier studies by *Cadigan* [1971], show that this fragmental material has undergone a great deal of alteration since deposition. Of particular importance, insofar as the present study is concerned, is the fact that hornblende, one of the most unstable iron-bearing silicate minerals, is generally absent from the present-day Moenkopi Formation. *Cadigan* [1971], who studied the regional distribution of heavy minerals in the Moenkopi throughout the Colorado Plateau, found hornblende at only one locality. We have found no hornblende in any of the samples we have studied.

From the arkosic nature of the detritus in the Moenkopi, however, it is reasonable to expect that hornblende was a common accessory mineral in the original sediments throughout the region, particularly along the eastern margin of the depositional basin, where the sediments were derived from

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Precambrian high-grade metamorphic and plutonic rocks in the Uncompahgre Highland. The roots of the same crystalline source rocks are presently exposed in the Uncompahgre Plateau, and the modern sediments derived from them are rich in hornblende. The Moenkopi sediments near the uplift, for example, those from the Salt Creek location, should have been similarly rich in hornblende at the time of deposition. It could be argued that the lack of hornblende in the Moenkopi Formation is evidence of extensive weathering during the Triassic in the source area prior to erosion. However, many writers [*Koldewijn,* 1958; *Nota,* 1958; *Van Andel and Postma,* 1954; *Walker,* 1974] indicate that at the surface of the earth today, even where weathering is intense (such as in tropical regions), hornblende and other unstable ferromagnesian silicate minerals are common constituents of sediments wherever these minerals are present in the source rocks. Moreover, the Moenkopi Formation contains thick beds of primary gypsum [*McKee,* 1954], suggesting that the regional climate at that time was either arid or semi-arid and that weathering of unstable minerals, therefore, would have been minimal.

Our contention that hornblende was initially present and has since been destroyed by intrastratal alteration is supported by the presence, in some samples, of intensely altered potassium feldspars (Figures 10*d* and 10*e*). The intensity and type of alteration vary from place to place. For example, X ray diffraction analyses of typically altered feldspars in the lower massive sandstone in the Wupatki Member at the Gray Mountain locality show that the daughter clay is kaolinite, whereas the daughter clay in altered feldspars from the Salt Creek locality is mixed-layer illite-smectite. Such differences reflect, on both local and regional scales, the inhomogeneity of the sediments, vagaries of groundwater movement through them, and the variability of the interstitial chemical environments. If relatively stable minerals such as potassium feldspar have been altered, one would expect that minerals such as hornblende, which under most near-surface conditions are much less stable than potassium feldspar, would be completely or nearly completely destroyed.

The idea that unstable minerals such as hornblende were important accessory minerals in the original deposits is further supported by the common occurrence of the grain-shaped dissolution voids, which are concentrated in heavy mineral laminae (Figures 7*c* and 7*d*). Inasmuch as these voids have the approximate size and shape of the associated heavy mineral grains, we believe that most of them are the result of complete dissolution of unstable iron-bearing silicate minerals.

That unstable mineral grains dissolve intrastratally over time is documented by studies of red beds of Cenozoic age in Baja California and southwestern United States [*Walker,* 1967, 1976; *Walker et al.,* 1978]. In those beds, dissolution of the minerals is incomplete, and relics of the partly dissolved grains still remain in the rocks.

The dissolution of unstable iron-bearing silicates from the Moenkopi sediments would have released ample amounts of iron ions into percolating groundwater, allowing the formation of ultrafine pigment, microcrystalline hematite crystals, and crystals of specular hematite, under the appropriate Eh-pH conditions. In addition, the breakdown or recrystallization of both detrital and authigenic clay minerals, as well as the alteration of biotite, would have provided other important sources of iron ions [*Carroll,* 1958; *Walker et al.,* 1967; *Walker and Honea,* 1969; *Walker,* 1976].

Petrographic observations also suggest that another source of iron ions was the in situ dissolution of detrital grains of magnetite and ilmenite. Many of the detrital opaque grains, particularly those of martite, have embayed margins and internal voids (Figures 7*c,* 9*a,* 9*e,* and 9*f*), indicating that portions of the grains have been dissolved. In addition, grains of ilmenite ($FeTiO_3$) have been altered to leucoxene (TiO_2), attesting to the release of iron ions. Dissolution of magnetite (martite) and ilmenite grains, most of which contain variable portions of titanium, can additionally provide a source of titanium for the growth of anatase and other titanium-bearing authigenic minerals that are found in association with the authigenic hematite species (Figures 6*a* and 7*e*).

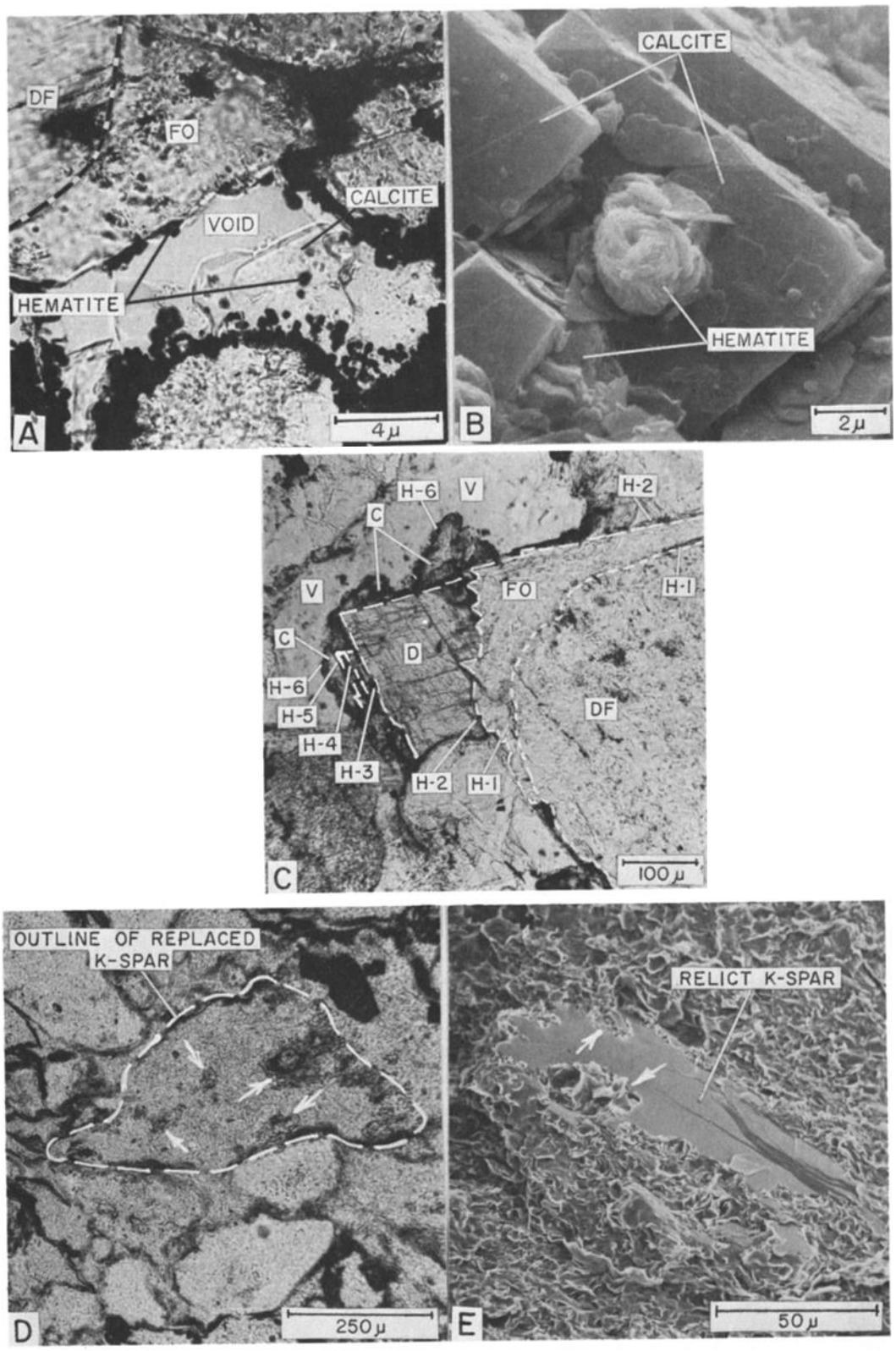
SUMMARY AND CONCLUSIONS

The results of our petrologic study show that the hematite in red beds in the Moenkopi Formation has a variety of forms and origins. There is unequivocal evidence that much of it is authigenic. Clearly authigenic varieties include microcrystalline translucent hematite and coarser crystals of black opaque specular hematite. Both forms have been precipitated in interstitial and secondary voids, in interstitial matrix, between cleavage lamellae in intrastratally altered biotite grains, and as encrustations on detrital grains. In addition, crystals of specular hematite form optically continuous overgrowths on hematitized magnetite grains (martite), and microcrystalline hematite has replaced earlier formed authigenic pyrite.

Specular hematite is also present as polycrystalline and monocrystalline grains of martite. The hematite in these grains could have originated as a result of oxidation in the source area, intrastratal oxidation, or more likely, a combination of both. Thermodynamic data concerning the chemical stability of magnetite and hematite, coupled with data concerning the relative abundance of magnetite and hematite in modern sediments, indicate that much, if not most, of the hematitization of the magnetite took place after deposition as the grains were bathed by groundwaters in the Eh-pH stability field of hematite. Some specular hematite is also present in grains of ilmenite that presumably have been oxidized intrastratally in a manner analogous to the martitization of the

Fig. 10. (Opposite) (*a*) Thin section photomicrograph showing spheroids of hematite within crystals of sparry calcite cement and coating the surface of an authigenic feldspar overgrowth (FO) developed on a detrital grain of feldspar (DF). Hematite (black) also is present as coatings on detrital grains (as in lower part of photo) and as interstitial cement. Ali Baba Member on east flank of Sinbad Valley near Gateway, Colorado. (*b*) SEM photomicrograph of same sample as in 10*a*, showing microrosettes composed of authigenic hematite crystals on the surface of and within crystals of authigenic calcite. (*c*) Thin section photomicrograph displaying a complex paragenetic sequence, explained in the text. DF, detrital feldspar; FO, feldspar overgrowth; D, dolomite; C, calcite: V, void; H-1 to H-6, successive stages of hematite and its precursor oxides. Transmitted ordinary light. Middle Moenkopi along Salt Creek, southwest of Gateway, Colorado. (*d*) Thin section photomicrograph showing relics of several pervasively replaced framework silicate grains. The outlined area shows the approximate boundary of a grain of potassium feldspar that has been extensively replaced by mixed-layer illite-smectite. The arrows show the locations of several optically continuous relics of the parent potassium feldspar (identified with the aid of cobaltinitrite stain). Transmitted ordinary light. Upper Moenkopi along Salt Creek, southwest of Gateway, Colorado. (*e*) SEM photomicrograph showing a relic of an altered grain of potassium feldspar analogous to the relics of feldspar remaining in the area outlined in 10*d*. Note the deep embayments (arrows) of the daughter clay around the periphery of the relic potassium feldspar. These embayments attest to the replacement origin of the clay. Same sample as in 10*d*.

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magnetite. Finally, a small amount of specular hematite is also present in grains composed of ilmenite-hematite intergrowths ('tiger striped' grains) and in monocrystalline grains containing rutile exsolution platelets. The latter two varieties are products of high-temperature processes and are the only forms of hematite of unquestionable detrital origin in the Moenkopi red beds.

All of these various forms of hematite are coarser grained than the superparamagnetic threshold size of hematite and, therefore, are capable of contributing to the remanence. The detrital tiger striped grains probably have a minimal effect, because they comprise a very small percentage of the total hematite population. Moreover, the grains are composed of about 25–75% nonmagnetic ilmenite, which makes alignment

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in the earth's field more difficult than if they were composed entirely of hematite. Similarly, the monocrystalline detrital grains containing rutile exsolution platelets are so relatively rare that they also probably contribute little to the total remanence.

The following observations constitute compelling evidence that the microcrystalline hematite and the crystals of specular hematite are authigenic and were precipitated in the rocks long after deposition of the host sediments: the occurrence of the hematite as euhedral crystals lining interstitial voids and replacing detrital silicate minerals and authigenic pyrite; its superimposed relationship on other authigenic minerals, the growth of which must have required considerable geologic time; and its occurrence in voids formed by dissolution of detrital grains, many of which probably were silicate minerals that dissolve slowly even on a geologic time scale.

The iron in the various forms of authigenic hematite as well as that in the ultrafine red pigment was likely derived almost entirely from intrastratal sources and probably mainly from alteration of iron-bearing detrital grains. The most likely source minerals are the easily altered detrital ferromagnesian silicates, such as hornblende and biotite, and detrital oxides, such as magnetite and ilmenite. However, all types of unstable iron-bearing grains, including detrital and authigenic clay minerals, probably have contributed some of the iron.

The Moenkopi red beds seem to be ancient analogues of red beds that are forming diagenetically today in deposits of Cenozoic age in southwestern United States and northwestern Mexico [*Walker*, 1967, 1976; *Walker et al.*, 1978]. They differ from these younger red beds mainly in that the Moenkopi red beds show effects of more advanced stages of intrastratal alteration. Feldspar grains, particularly relatively stable potassium feldspar, tend to have been more extensively altered in the Moenkopi red beds, and presumably, the more labile minerals such as hornblende have been nearly or completely removed. In addition, euhedral hematite crystals are more abundant in the Moenkopi red beds, with the maximum size of the crystals being many times larger than in the Cenozoic red beds. Moreover, red beds in the Moenkopi Formation generally are less brightly colored than those of Cenozoic age, and we believe the color difference may be related to differences in crystal size of the hematite. All these differences, in our opinion, reflect the difference in the length of time that intrastratal alterations have been in progress in the two sequences.

We wish to emphasize that our evidence showing that hematite is present in a variety of forms having diverse origins refutes the commonly held belief that hematite in red beds can be divided into only two groups: pigment, and grains of detrital specularite [see, e.g., *Collinson*, 1974, and *Elston and Purucker*, 1979]. Such a twofold division is an oversimplification of a highly complex association. It follows that interpretations concerning the time of emplacement of remanence that are based on a twofold division of hematite are of doubtful validity.

Similarly, because of the variety of hematite forms and sizes, the results of acid-leaching demagnetization experiments [*Roy and Park*, 1972; *Elston and Purucker*, 1979] should be viewed with caution, particularly if detailed petrographic studies did not accompany the experiments. Bleaching of the sample during the early experimental stages is obviously indicative of the removal of the color associated with the ultrafine pigment and finer microcrystalline hematite. Beyond that leaching level, the results of such studies become more difficult to assess. Large authigenic crystals and crystal aggregates can be just as resistant to leaching as detrital specularite grains, especially if the authigenic grains are encased in silicate grains (for example, hematite pods in biotite). Moreover, most detrital opaque grains consist principally of martite and to a lesser degree, of oxidized ilmenite, and the remanence they carry most likely reflects the field at the time of martitization and oxidation-diffusion, respectively, not the field at the time of deposition.

Finally, we maintain that any viable theory concerning the origin of remanence in the Moenkopi red beds must take into account the extensive mineralogic alterations that have prevailed since deposition and also the abundance of varietal forms of hematite in the rocks, most of which have formed authigenically.

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